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1. Title

Implementation of Text Cleaning Techniques in Natural Language Processing

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2. Aim

To study and implement **text cleaning techniques** to preprocess raw textual data for Natural Language Processing tasks.

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3. Objectives

- To understand the importance of text cleaning in NLP
- To remove unwanted elements such as punctuation, numbers, and special characters
- To convert text into a clean and standardized format
- To prepare cleaned text for further NLP processing

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4. Theory

4.1 Text Cleaning in NLP

Text cleaning is a preprocessing step used to remove **noise** from raw text data. Real-world text often contains unnecessary symbols, mixed casing, extra spaces, and irrelevant characters that can affect model performance.

4.2 Common Text Cleaning Operations (As per the File)

- **Lowercasing**
  Converts all characters into lowercase to maintain uniformity.

- **Removal of Punctuation**
  Eliminates symbols such as commas, periods, and special characters.

- **Removal of Numbers**
  Removes numeric values that may not contribute to textual meaning.

- **Removal of Extra Whitespaces**
  Ensures proper spacing between words.

4.3 Importance of Text Cleaning

- Improves accuracy of NLP models

- Reduces noise in textual data

- Helps in better feature extraction

- Acts as a base for tokenization, stemming, and vectorization

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5. Program Codes

Text cleaning:

1. HTML Tags Cleaning

[4]
```
sample = '<p>Browsers automatically format the q tag.</p><p>The goal remains: <q>Education leads to innovation.</q></p>'
```

[5]
```
import re
def striphtml(data):
  p = re.compile(r'<.*?>')
  return p.sub('', data)
```

[6]
```
striphtml(sample)
```

```
'Browsers automatically format the q tag.The goal remains: Education leads to innovation.'
```

2. Emoji Cleaning

[7]
```
emoji_text =  'Hi I am 😊''How are you 😊'
```

[8]
```
emoji_text.encode('utf-8')
```

```
b'Hi I am \xf0\x9f\x98\x8aHow are you \xf0\x9f\x98\x8a'
```

3. Spell Checker

[9] 5s

```
!pip install textblob
```

```
Requirement already satisfied: textblob in /usr/local/lib/python3.12/dist-packages (0.19.0)
Requirement already satisfied: nltk>=3.9 in /usr/local/lib/python3.12/dist-packages (from textblob) (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob) (4.67.1)
```

[10] 4s

```
from textblob import TextBlob
incorrect_text = 'certnia cnoditoins are moidfeid'
textblob = TextBlob(incorrect_text)
corrected_text = textblob.correct()

print("Original Text:", incorrect_text)
print("Corrected Text:", corrected_text)
```

```
Original Text: certnia cnoditoins are moidfeid
Corrected Text: certain conditions are modified
```

4. Removal of unwanted words and characters

[11] 0s

```
tweet = 'Your mental health matters as much as your physical health.  🏋️💚Take breaks, seek support, and prioritize self-care. Lets break the stigma and support
```

[12] 0s

```
tweet.lower()
```

```
'your mental health matters as much as your physical health.  🏋️💚take breaks, seek support, and prioritize self-care. lets break the stigma and support one ano
ther. @mentalhealth @wellnessforever #mentalhealthawareness #endthestigma'
```

[13] 0s

```
def clean_text(text):

    #lowering the text
    text = text.lower()

    #removing anything starts from @ #
    text = re.sub("[@|#]\w+","",text)

    return text

clean_text(tweet)
```

```
<>:7: SyntaxWarning: invalid escape sequence '\w'
<>:7: SyntaxWarning: invalid escape sequence '\w'
/tmp/ipython-input-3140161321.py:7: SyntaxWarning: invalid escape sequence '\w'
  text = re.sub("[@|#]\w+","",text)
'your mental health matters as much as your physical health.  🏋️💚take breaks, seek support, and prioritize self-care. lets break the stigma and support one ano
ther.    '
```

Install Regular Expression

[14] 9s

```
!apt-get install re
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  re
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 34.5 kB of archives.
After this operation, 108 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 re amd64 0.1-7build1 [34.5 kB]
Fetched 34.5 kB in 0s (78.2 kB/s)
Selecting previously unselected package re.
(Reading database ... 117540 files and directories currently installed.)
Preparing to unpack .../re_0.1-7build1_amd64.deb ...
Unpacking re (0.1-7build1) ...
Setting up re (0.1-7build1) ...
Processing triggers for man-db (2.10.2-1) ...
```

[15] 0s

```
def clean_text(text):

    #lowering the text
    text = text.lower()

    #removing emoji's
    text = text.encode('utf-8')

    return text

clean_text(tweet)
```

```
b'your mental health matters as much as your physical health.  \xf0\x9f\x8c\xb1\xf0\x9f\x92\x9atake breaks, seek support, and prioritize self-care. lets break
the stigma and support one another. @mentalhealth @wellnessforever #mentalhealthawareness #endthestigma'
```

Encode the emoji into utf-8 format and decode the same to ignore ascii characters

[16]

```python
import re
def clean_text(text):


    #lowering the text
    text = text.lower()

    #removing anything starts from @ #
    text = re.sub("[@|#]\w+","",text)

    #encoding emoji's
    text = text.encode('utf-8')
    text = text.decode('utf-8', 'ignore')


     #removing unicode text format
    text = re.sub(r'[^\x00-\x7F]+', '', text)
    return text

clean_text(tweet)
```

```
<>:9: SyntaxWarning: invalid escape sequence '\w'
<>:9: SyntaxWarning: invalid escape sequence '\w'
/tmp/ipython-input-1614530550.py:9: SyntaxWarning: invalid escape sequence '\w'
  text = re.sub("[@|#]\w+","",text)
'your mental health matters as much as your physical health.  take breaks, seek support, and prioritize self-care. lets break the stigma and support one another.
'
```

Removal of URLs

[17]

```python
import re
def remove_url(text):
  pattern = re.compile(r'https?://\S+|www\.\S+')
  return pattern.sub(r'', text)
```

[18]

```python
text1 = 'Appple services and Products https://www.apple.com'
text2 = 'Wesite builders used to improve domain https://websitebuilders.com'
text3 = 'Welcome to Google notebook www.google.com/notebook'
text4 = "Tokenization and lemmatization www.geeksforgeeks.org/nlp-how-tokenizing-text-sentence-words-works/"
```

[19]

```python
remove_url(text4)
```

```
'Tokenization and lemmatization '
```

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6. Result and Outcomes

Result

- The raw input text was successfully cleaned.
- Punctuation marks, numbers, and special characters were removed.
- The final output text was uniform, readable, and noise-free.

Outcomes

- Understood the necessity of text cleaning in NLP
- Learned different cleaning techniques used in preprocessing
- Gained practical experience using Python for text preprocessing
- Prepared text data for further NLP tasks

7. Conclusion

Text cleaning is an essential preprocessing step in Natural Language Processing. It helps eliminate unwanted noise and ensures consistency in textual data. Through this experiment, various text cleaning techniques were successfully implemented, demonstrating their importance in improving the quality and effectiveness of NLP applications